

Functional and Non-Functional Requirements Specification

Chat Application (WebSocket Programming) — Team 5, Section K

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1. Functional Requirements

ID	Requirement	Description	Acceptance Criteria — Pass	Acceptance Criteria — Fail	Priority	Rationale
FR-01	User Registration	Allow a new user to create a unique account using email, username, and password.	Account is created when a unique username/email and a password meeting complexity policy are submitted; confirmation is shown and the user can proceed to log in.	Registration is rejected with a clear error when the email/username is already in use, a required field is missing, or the password fails the complexity policy.	High	Registration is the entry point for all other features; without it no user can access the application.
FR-02	User Authentication	Allow a registered user to log in securely using username/password.	User enters correct credentials and is granted access to their account, contacts, and chat history within the expected response time.	Login is denied with an appropriate error for incorrect credentials, and the account is throttled or locked after repeated failed attempts.	High	Secure login protects user accounts and is required before any messaging feature can be used.
FR-03	Session Management	Maintain a user's logged-in state across application sessions and app restarts.	User closes and reopens the app within the session validity window and is returned to their account without re-entering credentials.	An expired or invalid session forces the user back to the login screen instead of granting silent access.	Medium	Reduces login friction and improves usability for returning users.
FR-04	Password Recovery	Provide a secure mechanism for users to reset a forgotten password via email.	User requests a reset, receives a time-limited email link, sets a new password, and can log in with the new credentials.	An expired or already-used reset link is rejected, and no password change occurs without a valid, verified link.	High	Users must be able to regain account access without contacting support, while preventing unauthorized resets.
FR-05	Profile Management	Allow users to create and update their profile picture, display name, and status message/bio.	Changes to picture, display name, or status are saved and immediately visible to the user and their contacts.	Invalid input, such as an oversized image or disallowed characters, is rejected with an explanatory message and the profile remains unchanged.	Medium	Personalization increases engagement and helps contacts recognize one another.
FR-06	Contact/Friend Management	Allow users to add, remove, and manage their contacts via friend requests.	A sent friend request appears for the recipient, who can accept (both users become contacts) or decline (no relationship is created).	Duplicate or self-requests are blocked, and a removed contact no longer appears in either user's contact list.	High	A contact list is a prerequisite for initiating any one-to-one or group chat.

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FR-07	One-to-One Chat Initiation	Allow a user to start a private conversation with a contact from their contact list.	Selecting a contact opens a chat window ready for message entry, reusing an existing conversation thread if one already exists.	Attempting to open a chat with a blocked user, or a non-contact where restricted, is prevented with a clear message.	High	Core capability required to deliver the application's primary purpose of instant messaging.
FR-08	Send Text Message	Allow a user to compose and send a text-based message in an open chat.	Typed text is transmitted when the send action is triggered and appears in the sender's chat view immediately.	Empty messages are not sent, and messages exceeding the maximum allowed length are rejected with a warning.	High	Sending text is the fundamental action of a messaging application.
FR-09	Receive Text Message	Receive and display incoming messages to the user in real time.	A message sent by a contact appears in the recipient's open chat within the expected latency, without a manual refresh.	A message is never lost, duplicated, or displayed in the wrong conversation.	High	Real-time receipt is essential to the instant-messaging experience the application promises.
FR-10	Message Delivery Status	Provide visual feedback on message status: sending, sent, and delivered.	The status icon transitions from a single checkmark (sent to server) to a double checkmark once the recipient's device confirms receipt.	Status remains at "sending" and an error or retry option is shown if delivery cannot be confirmed within the timeout period.	Medium	Gives senders confidence that a message reached its destination.
FR-11	Read Receipts	Provide a visual indicator when a recipient has read a message.	Delivery checkmarks turn blue only after the recipient has opened the chat and viewed the message.	Checkmarks do not turn blue if the recipient has disabled read receipts or has not yet opened the chat.	Low	Provides social confirmation of engagement but is not critical to core delivery.
FR-12	Message History/Persistence	Store and retrieve the entire conversation history for any chat.	Scrolling up in a chat window loads earlier messages in correct chronological order, including after logout and re-login.	No message is permanently lost, and retrieval never returns messages out of order or from the wrong chat.	High	Users expect to reference past conversations at any time.
FR-13	Typing Indicators	Display a real-time indicator when the other user in a chat is actively typing.	The indicator appears within a short delay after the other user starts typing and disappears once they stop or send the message.	The indicator does not appear if the typing user has disabled it, and it never persists after the message is sent.	Low	Enhances conversational flow but is not essential to message delivery.
FR-14	Group Chat Creation	Allow a user to create a new chat group by selecting multiple contacts.	Selecting two or more contacts and confirming creates a group chat visible to all selected members.	Group creation fails gracefully with an error if fewer than the minimum required number of members is selected.	Medium	Group messaging is a widely expected feature for team, family, and community communication.
FR-15	Group Chat Management	Allow group admins to add or remove members, change the group name/icon, and assign admin roles.	An admin's changes to membership, name, icon, or roles are applied and become visible to all group members.	A non-admin member attempting an admin-only action is blocked and shown a permissions error.	Medium	Keeps group chats organized and moderated as membership changes over time.

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FR-16	Message Edit	Allow the sender of a message to modify its content after it has been sent.	The edited message displays the updated text with an "edited" label, visible to all participants.	A user cannot edit another user's message, and edits attempted after the allowed time window are rejected.	Low	Lets users correct mistakes without deleting and resending a message.
FR-17	Message Deletion	Allow the sender of a message to delete it for themselves or for everyone in the chat.	"Delete for everyone" removes the message from all participants' views; "delete for me" removes it only from the sender's view.	A user cannot delete another user's message, and deletion after the allowed time window, if defined, is blocked.	Medium	Supports user privacy and correction of mistakenly sent content.
FR-18	File/Image Sharing	Allow users to send and receive various file types, including images, videos, and documents.	A supported file type under the size limit uploads, sends, and is viewable or downloadable by the recipient.	Unsupported file types or files exceeding the size limit are rejected with a clear error before upload begins.	Medium	Media sharing is a standard expectation of modern chat applications.
FR-19	Push Notifications	Send a notification to a user's device for new messages when the app is in the background or closed.	A notification with a message preview and sound/vibration appears within the expected delay when a new message arrives.	No notification is shown for muted chats or contacts, and no message triggers a duplicate notification.	High	Keeps users engaged and responsive without requiring the app to remain open at all times.
FR-20	Search Functionality	Allow users to search for specific text within chat history or search for contacts/groups.	A search query returns all matching messages, contacts, or groups, clearly highlighted, within an acceptable response time.	A query with no matches returns a clearly labeled empty result rather than an error or stale data.	Medium	Helps users quickly locate past conversations or people as chat history grows.
FR-21	Reconnection Handling	Automatically attempt to reconnect to the server after a temporary network disruption.	On network restoration, the client automatically reconnects and synchronizes any missed messages without user action.	Queued outgoing messages are not lost during a disruption, and no manual app restart is required to resume messaging.	High	Mobile and network conditions are unreliable; automatic recovery is essential for a real-time application.
FR-22	Block/Mute/Report Users	Provide users with tools to block, mute notifications from, or report other users for misconduct.	A blocked user can no longer send messages to the blocking user, and a submitted report is logged for review.	A blocked user is not notified that they have been blocked, and messages from a blocked user are not delivered.	Medium	Protects users from harassment and abuse, supporting overall trust and safety.
FR-23	Message Reactions/Emojis	Allow a user to add an emoji reaction to a specific message.	Selecting a reaction attaches it to the message and is visible to all chat participants in real time.	Selecting the same reaction a second time toggles it off rather than stacking duplicate reactions.	Low	Provides a lightweight way to acknowledge messages without sending a full reply.
FR-24	Message Threads/Replies	Allow a user to reply directly to a specific message, creating a threaded context.	A reply displays a quoted reference to the original message, and selecting it navigates back to that message.	A reply to a message that was later deleted still shows a placeholder rather than causing an error.	Low	Reduces confusion in busy group chats by preserving conversational context.

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FR-25	User Mentions (@username)	Allow a user to mention other users in a group chat to send them a specific notification.	Typing "@" and selecting a member tags them in the message, and the mentioned user receives a distinct notification.	Mentioning a user who is not a member of the group does not resolve to a valid, notifiable tag.	Low	Helps direct attention to specific members in high-volume group conversations.
FR-26	Multi-device Synchronization	Synchronize a user's chat history and account status across all logged-in devices in real time.	A message sent or read on one device appears identically, with matching status, on all other logged-in devices.	A message read on one device is never shown as unread on another device once synchronization completes.	Medium	Users expect a seamless experience when switching between phone, tablet, and desktop.
FR-27	Voice Messages	Allow users to record and send short audio clips.	A recorded clip within the allowed duration uploads, sends, and is playable by the recipient with a visible duration indicator.	Recording is prevented without microphone permission, and clips exceeding the duration limit are rejected.	Low	Offers a convenient alternative to typing, particularly on mobile devices.
FR-28	Voice/Video Calls	Facilitate real-time, one-to-one or group voice and video calls between users.	A call connects within the expected setup time, with audio and, where applicable, video flowing in both directions.	A declined or unanswered call ends gracefully without hanging the app or leaving open network connections.	Medium	Extends the application beyond text into richer, real-time communication.
FR-29	Stickers/GIFs Integration	Provide an interface for users to search for and send stickers or animated GIFs.	A searched sticker or GIF sends and renders correctly, including animation, in the chat for all participants.	A search with no results returns an empty state rather than an error, and unsupported formats are not sent.	Low	Adds expressive, engaging content options favored by many chat users.

2. Non-Functional Requirements

The table below lists all non-functional (quality attribute) requirements, each with a description, pass/fail acceptance criteria, priority, and rationale for inclusion.

ID	Requirement	Description	Acceptance Criteria — Pass	Acceptance Criteria — Fail	Priority	Rationale
NFR-01	Performance / Latency	Deliver a message from sender to recipient in under 200ms on a stable network connection.	Measured end-to-end delivery latency is at or below 200ms for at least 95% of messages under normal network load.	Latency that consistently exceeds 200ms under stable network conditions is treated as a defect requiring investigation.	High	Low latency is central to the real-time value proposition of the application.
NFR-02	Scalability	Architecture must support 100,000 concurrent users with no degradation in performance.	Load testing at 100,000 concurrent connections shows response times and delivery rates within defined targets.	A noticeable increase in latency, dropped connections, or errors occurring below the 100,000-user threshold fails this requirement.	High	The application must remain usable during growth and peak traffic periods.
NFR-03	Availability / Uptime	The system shall be operational and available to users at all times.	Measured uptime meets or exceeds the agreed service level (e.g., 99.9%) over a rolling measurement period.	Unplanned downtime that exceeds the allowed error budget within a measurement period fails this requirement.	High	Users rely on the application for ongoing communication; extended outages directly harm trust and usage.
NFR-04	Security	All message content shall be protected using end-to-end (E2E) encryption.	Independent security review confirms only sender and recipient devices can decrypt message content, with server storage limited to ciphertext.	Any scenario in which server administrators or a network intercept can read plaintext message content fails this requirement.	High	Confidentiality of private conversations is a core user expectation and often a regulatory requirement.
NFR-05	Data Integrity	The system must guarantee that messages are never corrupted, lost, or delivered out of their original order within a session.	A rapid sequence of messages is received by the recipient in the exact order sent, with content unchanged.	Any reordering, duplication, or corruption of message content within a session fails this requirement.	High	Conversations become confusing or unusable if message order or content cannot be trusted.
NFR-06	Robustness / Fault Tolerance	The system should handle unexpected errors or server failures gracefully without losing user data.	Simulated failure of a single server node results in automatic failover with no dropped messages or visible disruption to users.	A single node or service failure that causes data loss or requires manual intervention to restore service fails this requirement.	High	A distributed real-time system must tolerate partial failures without impacting the user experience.
NFR-07	Reliability	The system must successfully deliver all sent messages to the server.	Under normal operating conditions, 100% of client-submitted messages are acknowledged and stored by the server.	Any message accepted by the client but silently dropped before reaching the server fails this requirement.	High	Message loss undermines the fundamental trust required for a communication tool.

ID	Requirement	Description	Acceptance Criteria — Pass	Acceptance Criteria — Fail	Priority	Rationale
NFR-08	Ease of Use / Usability	A new user should be able to register, find a contact, and send their first message in under 60 seconds without any training.	Usability testing shows first-time users completing registration through first message send within 60 seconds, unaided.	Users who require external help or documentation, or who exceed the time target during testing, fail this requirement.	Medium	Low onboarding friction directly drives adoption and reduces support burden.
NFR-09	Portability	A user can switch between their iPhone, Android tablet, and Windows PC and have a consistent experience.	Core features and UI behave consistently, and account state is preserved, across all supported platforms.	A feature that is missing, broken, or behaves inconsistently on a supported platform fails this requirement.	Medium	Users expect to move between devices without relearning the application or losing functionality.
NFR-10	Maintainability	The system's code shall be modular, well-documented, and have high test coverage to allow for easy updates and bug fixes.	Code review confirms a modular architecture, adequate documentation, and automated test coverage at or above the agreed threshold.	Tightly coupled modules, missing documentation, or coverage below the agreed threshold fails this requirement.	Medium	Reduces the cost and risk of future fixes, such as isolating the notification module from the chat core.
NFR-11	Logging	Log all critical errors, security events, and system performance metrics on the server side for diagnostics.	A sampled critical error, security event, or performance metric is verifiably captured in server logs with timestamp and context.	Any critical error or security event that occurs without a corresponding log entry fails this requirement.	Medium	Enables diagnosis of issues such as reported login failures, and supports security auditing.
NFR-12	Disaster Recovery	The system must be fully restorable from backups.	A recovery drill restores the system from backup to a fully functional state within the defined recovery time objective.	Data loss beyond the defined recovery point objective, or failure to restore within the recovery time objective, fails this requirement.	High	Protects the business and its users against catastrophic data loss events.